INCH-POUND

MIL-PRF-1/964D
27 August 1999
SUPERSEDING
MIL-E-1/964C
30 June 1978

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING
TYPE 6378

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: TR, tunable, frequency range 8.49 to 9.6 GHz.

ABSOLUTE RATINGS:

Parameter: Unit:	po (transmitter) kW	Ebb (open circuit) V dc	Du ---	Ii µA dc	Alt ft	TA °C
Maximum: Minimum:	30 ---	--- -500	0.001 ---	200 100	10,000 ---	+100 -40

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1.
Weight: 5 ounces (approximately).

TEST CONDITIONS:

Parameter: Unit: Tolerance:	Ebb V dc ---	Ii µA dc ---	po kW ±10%	tp µs ±10%	Du --- ---	Ri MΩ ---
Test 1:	-500	---	---	---	---	1.25
Test 2:	---	100	---	---	---	1.25
Test 3:	-500	---	30	1.0	0.001	1.25
Test 4:	---	100	30	0.5	0.0005	1.25
Test 5:	---	100	10	0.5	0.0005	1.25

Test frequencies	
F1	8.490 GHz ± 0.1 %
F2	9.375 GHz ± 0.1 %
F3	9.600 GHz ± 0.1 %

See footnotes at end of table I.

GENERAL:

Qualification: Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Testing and inspection.

Requirement or test	Method	Test	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 1			1/				
Ignitor ignition time	4401	1		t	---	1.0	sec
Ignitor voltage drop	4406	2		Eid	200	400	V dc
Insertion loss (1) (integral cavity)	4416	---	F = F2	Li	---	1.5	dB
Ignitor interaction (insertion loss)	4421	2		ΔLi	---	0.2	dB
Tuning range	4426	---	F = F1 to F3	---	5	---	turns
Leakage power	4452	5	F = F2, T = 25°C	$\frac{Po}{Du}$	---	30	mW
Temperature cycling (nonoperating)	1027	---		---	---	---	---
Conformance inspection, part 2							
High-frequency vibration	1031	---	t = 15 minutes	---	---	---	---
Loaded Q	4461	---		QL	---	350	---
Vibration detuning	4016	---	F = F2, t = 12 hours	ΔF	---	±3	MHz
Ignitor oscillation (obsolescent)	4411	---		Li	---	60	μA dc
Insertion loss (2) (integral cavity)	4416	---	F = F1 and F3	Li	---	1.5	dB
Temperature coefficient of frequency	4466	---	F = F1 2/	ΔF	---	-20	MHz
Recovery time	4471	4	F = F2	t	---	4.0	μs
Pressurizing	4002	---	45 lb_f/in^2	---	---	---	---

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Requirement or test	Method	Test	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 3							
Life test	4551	3	F = F2, Group D	t	500	---	hrs
Life test end points:	---						
Ignitor voltage drop	4406	2		Eid	---	450	V dc
Insertion loss (1)	4416	---	F = F2	Li	---	2.0	dB
Ignitor interaction	4421	2		Li	---	0.5	mW
Leakage power	4452	5	F = F2, T = 25°C	$\frac{Po}{Du}$	---	35	mW
Recovery time	4471	4	F = F2	t	---	8.0	µs
Temperature cycling	1027	---	Group D	---	10	---	cycles

1/ Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be 1.0 percent, inspection level II.

2/ The frequency drift shall be measured with no adjustment of the tuning mechanism.

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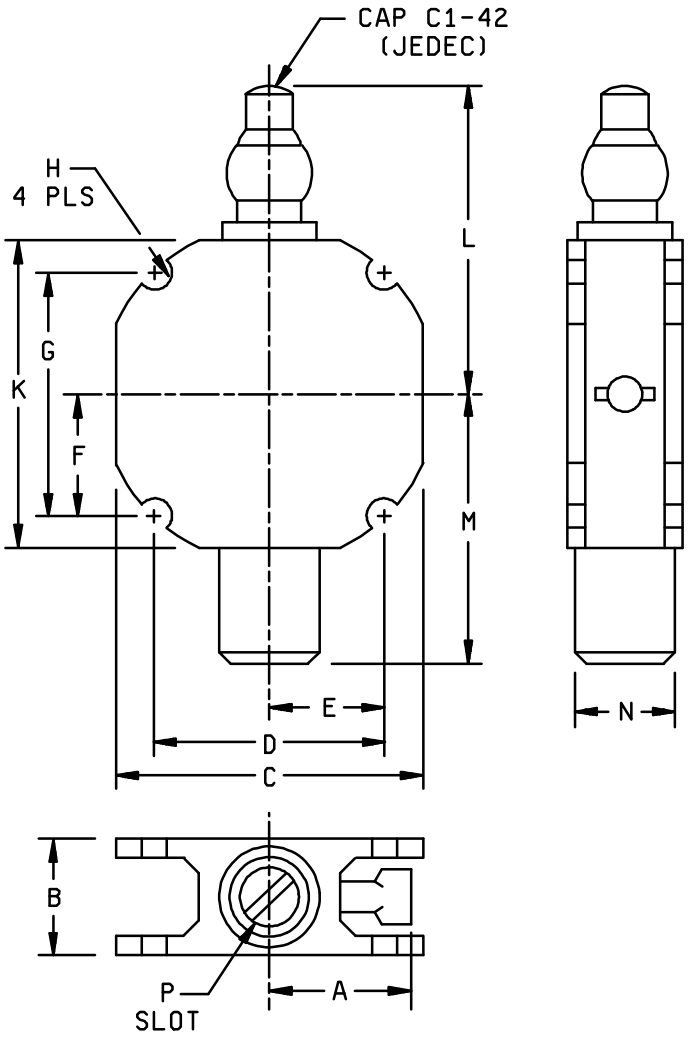

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 1				
A	---	.750	---	19.05
B	.607	.617	15.42	15.67
L	---	1.625	---	41.28
Conformance inspection, part 2				
D	1.216	1.224	30.89	31.09
G	1.276	1.284	32.41	32.61
H	.085 R	.095 R	2.16 R	2.41 R
M	---	1.422	---	36.12
N	---	.530 DIA	---	13.46 DIA
Conformance inspection, part 3				
C	1.610	1.640	40.89	41.66
E	.608	.612	15.44	15.54
F	.638	.642	16.21	16.31
K	1.610	1.640	40.89	41.66
P	.031	.062	0.79	1.57

NOTE: Exhaust tube shall not extend beyond flange.

FIGURE 1. Outline drawing of electron tube type 6378.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR

Preparing activity:
DLA - CC

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